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Why the PEA bill and the inverter app don't agree

Featured story · Two Solar ESS plants, fifteen hypotheses, one method · Rev.01, 7 November 2026

Summary

Owners often expect the units bought from the grid on their PEA bill to match the “grid import” in their inverter app. In October 2026 Solar ESS found two plants where the bill was **2.5 times** and up to **6 times** higher than the app. In both cases the solar panels were working normally. We tested fifteen possible causes. The PEA meter is the billing instrument and is almost always right; the app only knows what its current sensor (CT) sees. A constant ratio in both directions points to a CT scaling or wiring fault; a ratio that keeps growing points to loads the CT cannot see.

1. Two meters, two different jobs

Every grid-connected solar site has two devices counting energy at the grid connection. The **PEA meter** is a sealed, certified revenue meter. It counts every unit that passes through it, and the bill is based on it. The **inverter's CT or smart meter** is a control sensor. The inverter uses it to decide when to charge or discharge the battery and how much to export, and the app turns its readings into “grid import”, “export” and “load”.

When the two disagree by a few percent, that is normal. When they disagree by two times or more, the CT is not seeing what the PEA meter sees, and the app's “load” and “savings” figures are wrong too, because they are calculated from the CT.

2. Case A — a 5 kWp hybrid home near Bangkok

A 5.08 kWp hybrid system with a battery. From April to October 2026 the panels produced 6,466 kWh, a normal figure for this size. Over the same months the PEA bills showed 4,458 units bought, while the app recorded only 1,765 kWh.

Month 2026	Solar (kWh)	App: bought (kWh)	PEA bill (units)	PEA ÷ app
Apr	1,023.5	257.5	678	2.63×
May	983.7	318.9	768	2.41×
Jun	909.8	211.3	545	2.58×
Jul	912.0	241.2	606	2.51×
Aug	777.6	277.7	679	2.45×
Sep	812.8	201.0	549	2.73×
Oct	1,046.4	257.0	633	2.46×
Total	**6,465.8**	**1,764.6**	**4,458**	**2.53×**

Two details matter. First, the ratio is almost constant, between 2.4 and 2.7 times every month. Second, the export side shows the same ratio: in October the PEA meter counted 147 units sold, the app 62.5 kWh (2.4 times). The PEA reads the meter on the 1st of the next month, so bill months line up with app months.

3. Case B — a 22.5 kWp chicken farm in Chiang Rai

An on-grid system with no battery. The panels produced 8,451 kWh in the five months with complete data. The PEA bills showed 8,944 units bought; the app showed 2,290 kWh.

Month 2026	Solar (kWh)	App: total load (kWh)	App: bought (kWh)	PEA bill (units)	PEA ÷ app
May	2,392.5	2,849.9	503.2	1,408	2.80×
Jun	1,909.8	2,418.7	549.1	1,709	3.11×
Jul	1,188.4	1,477.7	326.7	1,171	3.58×
Aug*	450.7	557.7	120.8	1,404	—
Sep	1,462.4	1,915.2	493.3	2,094	4.24×
Oct	1,497.6	1,854.3	417.4	2,562	6.14×

* August has gaps in the monitoring data and is left out. Here the ratio is not constant: it grows from 2.8 to 6.1 times. In September and October the PEA bill is higher than the *entire* load the app recorded, solar included. From September to October the bill rose 22% while the app's grid import fell 15%. Export is only 40–60 kWh a month, far too small to explain a gap of 850–2,150 units.

4. Fifteen hypotheses, tested against the two cases

Each possible cause leaves a different fingerprint in the numbers. We listed every cause we have met in the field and asked: does this pattern match what we see?

#	Hypothesis	Fingerprint in the numbers	Case A	Case B
H1	Loads wired before the CT	Bill > app; gap grows when those loads run more; bill can exceed the app's total load	Unlikely: export shows the same ratio	**Strong**
H2	CT on a sub-board, not on the main incomer	Same as H1: whole buildings or circuits invisible	Unlikely	**Strong**
H3	One CT on a three-phase supply	App sees roughly one phase; gap moves with how load is spread over the phases	To check	**Strong**
H4	Wrong CT ratio or meter type set in the inverter	Constant ratio, the same for import and export	**Strong**	Unlikely: ratio not constant
H5	CT around only some of the parallel conductors	Constant ratio (e.g. one of two or three cables measured), both directions	**Strong**	Possible
H6	CT arrow reversed	Import shows as export and the other way round; battery and export control behave oddly	Ruled out: both directions positive	Ruled out
H7	CT on the wrong phase relative to the inverter	Power reads low or even negative because voltage and current are out of step	Possible	Possible
H8	CT jaw not fully closed, damaged or a long, poor extension	Readings low and unsteady; can drift over time	Possible	Possible

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#	Hypothesis	Fingerprint in the numbers	Case A	Case B
H9	CT clamped around both line and neutral	Currents cancel; app shows almost no grid exchange	Unlikely: app still sees ~40%	Unlikely
H10	Monitoring data gaps (dongle or Wi-Fi offline)	Missing days; solar also drops sharply in the same month	Ruled out: solar data complete	Only August (excluded)
H11	Different reading dates (bill month vs calendar month)	Shifts units between months, but totals over several months agree	Ruled out: read on the 1st	Ruled out: shifted months still 2–5×
H12	PEA meter multiplier (CT-operated meter on large supplies)	Bill = dial reading × ratio; if a reading is taken without the multiplier, units look small	Not applicable	To check on site
H13	Phase netting: the app sums phases, the PEA meter may not	On three-phase sites, export on one phase and import on another cancel in the app but can both be counted by the meter	Not applicable	Small: export only 40–60 kWh
H14	Estimated reading or PEA meter fault	A single odd month, later corrected; meter accuracy is about ±1–2%	Ruled out: seven steady months	Ruled out
H15	Another meter, another building or a shared supply	The bill being compared covers more than the solar system's switchboard	To check	To check

5. What the fingerprints tell us

> **Constant ratio, both directions → the CT measures a fixed fraction.** Case A: the CT ratio set in the inverter, or a CT around only part of the conductors (H4, H5). The meter sees all the current; the CT sees about 40% of it.

> **Growing ratio, bill above the app's total load → loads outside the CT.** Case B: circuits or whole houses wired before the CT, a CT on a sub-board, or one CT on a three-phase supply (H1, H2, H3). The new or seasonal load is on a circuit the CT does not measure.

In both cases the solar panels are not the problem. The problem is what the inverter can see, and that matters: it decides battery charging, discharge and export from the CT. In Case A the inverter sees only about 40% of the purchases, so the battery may sit idle while the house still buys from the grid. Using the PEA figures, Case A's real October use was about 1,480 kWh, not the 1,189 kWh in the app, and solar plus battery covered about 57% of it. In Case B the farm's real October use was about 4,000 kWh, not the 1,854 kWh the app showed.

6. How to check your own system

- **Compare every month.** Put the PEA “units bought” next to the app's “grid import” for the same period. Up to about 10% difference is normal.
- **Compare export too.** If export shows the same ratio as import, suspect CT scaling (H4, H5). If only import is off, suspect loads outside the CT (H1–H3).
- **Do a 24-hour test.** Write down the PEA meter reading and the app's grid import at the same moment, then again 24 hours later. Both should rise by nearly the same amount.
- **Have your installer check on site:** CT on the main incomer right after the PEA meter; around every parallel conductor of one phase only; arrow toward the house; one CT per phase on three-phase supplies; CT ratio and meter type set correctly; clamp-meter reading matching the inverter's grid power.

- **Check again after one week.** Once the CT sees everything, the inverter can also supply those loads from solar, and the bill should fall.

7. What Solar ESS does now

These two plants were found in the same month, which tells us the problem is not rare. Solar ESS now compares the PEA bill with the monitoring data every month for every plant it looks after, and sends an installer to check the CT whenever the two differ by more than 10%. Results after the site checks will be published in the Monthly Report.

This is a desk study based on PEA bills and monitoring exports; the causes are confirmed only by measurement on site. Site names, bill numbers and amounts are removed.

Solar ESS Working Group